

Nathan Ko

✉ n22ko@uwaterloo.ca  nathanko  nathanko378  nathanko.ca

EDUCATION

University of Waterloo

Waterloo, ON

Honours Mathematics, Financial Analysis and Risk Management - Major Average: 85%

Expected 2030

- Relevant Coursework: Linear Algebra, Algebra & Proof Techniques, Algorithms & Data Abstraction, Functional Programming, Financial Markets, Microeconomics
- Activities/Awards: Varsity Squash Team Captain, Waterloo President's Scholarship

Rockridge Secondary School

West Vancouver, BC

Top 6 Average: 97%

Graduated 2025

- Relevant Coursework: AP Statistics (5/5), Calculus 12 (100%), Computer Science 12 (100%), Economics 12 (95%)
- Activities/Awards: Peak Squash National Ranking #10, 2x Waterloo Math Contest School Champion + Certificate of Distinction, Weightlifting Club Executive, Rockridge Academic Scholarship, BC Achievement Scholarship

EXPERIENCE

Engineering Intern

Jul 2025 – Aug 2025

CYC Motor Ltd

Hong Kong

- Built and deployed an AI-powered customer support chatbot using a retrieval-augmented generation (RAG) pipeline over internal product manuals, FAQs, and support documentation.
- Designed prompt and persona engineering logic to control response tone, reduce hallucinations, and ensure grounded answers from LLMs.
- Integrated the chatbot into the company website, serving 2,000+ monthly user interactions, deflecting a significant portion of human support tickets.
- Built a B2B sales analytics dashboard in Excel to analyze revenue by region, reseller, and time period for management reporting.

Founder

Sept 2024 – Jun 2025

Aim High Tutoring

West Vancouver, BC

- Founded and scaled a tutoring business to 17 clients within 9 months, generating \$10K+ CAD in revenue.
- Managed assistant tutors, marketing, scheduling, finances, and client on-boarding, while designing individualized curricula improving academic performance for students in Grades 7–12.

Volunteer Squash Coach

Sept 2023 – Jun 2024

Evergreen Squash Club

West Vancouver, BC

- Coached and mentored 20+ junior athletes across 70 hours, creating structured training plans and performance tracking.
- Delivered training focused on skill development, physical strengthening, and competitive performance.

PROJECTS

Varsity Squash Training Scheduler | *Next.js, React, Supabase, OpenAI API*

Oct 2025 – Present

- Collaborated in a team to build a full-stack scheduling platform, matching our varsity squash players into training sessions based on availability and skill ratings.
- Designed and implemented the PostgreSQL data model and AI-driven scheduling pipeline using Supabase and the OpenAI API to generate and manage optimized training sessions.

Tech Layoffs vs Stock Impact Analysis | *Python, Pandas, Streamlit, Altair*

Aug 2025 – Nov 2025

- Developed a Streamlit Dashboard, analysing and visualising the correlation between tech layoffs and stock impact.
- Cleaned and merged 4,000+ global layoff records with Yahoo Finance market data using Pandas, performing correlation analysis and linear regression.

SKILLS AND INTERESTS

Languages: Python, C/C++, SQL, JavaScript/TypeScript, HTML, CSS, Racket

Frameworks & Tools: Pandas, NumPy, Flask, React, Node.js, Git, VS Code, Pycharm, Excel

Spoken Languages: English, Cantonese, Mandarin

Hobbies/Interests: Squash, Golf, Weightlifting, Chess, Poker, Ping Pong, Brazilian Jiu Jitsu, Cliff Jumping